Microsoft Elevate

CAPSTONE PROJECT

HOSPITAL PATIENT'S RECORDS

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

A lot of information is gathered by hospitals and healthcare organizations on a daily basis, including demographic information, personal information, and location information. Unfortunately, this information is often available in raw data forms such as spreadsheets or databases, and it is often hard for administrators and analysts to comprehend crucial information in terms of patterns and trends. Analysing such large amounts of information may take a lot of time and may lead to missing out on crucial information.

The main objective of this project is to transform raw patient data into meaningful and easy-to-understand visual information. By using Microsoft Power BI Desktop, this project will develop an interactive dashboard that displays patient data through charts, graphs, maps, and tables. These visualizations will help users analyse key aspects such as gender distribution, marital status, race, ethnicity, and geographic location of patients.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of predicting the required bike count at each hour to ensure a stable supply of rental bikes. This involves leveraging data analytics and machine learning techniques to forecast demand patterns accurately. The solution will consist of the following components:

Data Collection:

The dataset used in this project was downloaded from the website "Analytics Maven"[1]. This website provides sample datasets for practice in data analysis. The patient dataset was downloaded and imported into Microsoft Power BI Desktop. It contains information such as demographic and location details, which are used for creating visualizations.

Data Processing:

The collected dataset is then processed with the help of the Microsoft Power BI Desktop application. Data cleaning is done with the help of Power Query. In this process, duplicate values, missing values, and data consistency are handled. In this case, the columns were formatted appropriately, and the unwanted columns were removed. After this process, the data is ready for analysis with the help of the Power BI dashboard.

PROPOSED SOLUTION:

Data Analysis:

Microsoft Power BI Desktop is used to analyze the data. With the help of this tool, the patient data is analyzed, and important trends are identified. Different types of visualizations, including charts, maps, and tables, are used to analyze the gender distribution, marital status, race, ethnicity, and geographic location of the patients.

Data Visualization:

For the purpose of data visualization, the application used was Microsoft Power BI Desktop, which helped in presenting the analyzed data in an easily understandable format. Various types of visualizations such as bar charts, pie charts, maps, and tables have been created in order to understand the distribution of patients based on their demographic details and geographical locations.

SYSTEM APPROACH:

Hardware Requirements:

- Processor: Intel i3 or Higher
- RAM: Minimum 4 GB
- Storage: 8 GB free disk space

Software Requirements:

- Operating System: Windows 10 or above
- Tool Used: Microsoft Power BI

Data Source:

- Hospital Patient Data set in CSV format
- The data source for the current project includes the data related to patients, which is downloaded from the website Analytics Maven.

Libraries/Tools Required to build the Model:

Power Bi Desktop:

The main application used to develop the analytical dashboard is Microsoft Power BI Desktop. It facilitates the import of data, creation of a data model, and development of interactive visualization. Using this application, several types of charts like line charts, KPI cards, donut charts, and bar charts were developed using the drag-and-drop feature to effectively visualize the crop production.

Power Query Editor:

The Power Query Editor is used for data transformation and preprocessing. This function helps to clean the dataset by removing duplicate values, changing data types, and handling missing values. This process ensures that the dataset is properly structured before analysis.

DAX(Data Analysis Expressions)

DAX is used to create calculated measures and perform data calculations. Measures like total patient, Gender, and t area can be created using DAX formulas. These data calculations are helpful in deriving meaningful insights and improving the analytical capabilities of the dashboard.

ALGORITHM & DEPLOYMENT:

1. Algorithm section:

For the purpose of this project, data analysis and visualization techniques have been employed using Microsoft Power BI Desktop. Instead of relying on sophisticated machine learning algorithms, basic data analysis techniques like data aggregation, filtering, and grouping have been employed. These techniques can be used to obtain significant insights regarding patients in an effective way.

Data Input:

The algorithm processes Hospital patient's data with the following features:

1. Full name
2. Birthdate
3. Death Date
4. Marital
5. Gender
6. Birthplace
7. Address
8. City
9. States
10. Country
11. 1.2 Zip
12. 1.2 Lat
13. 1.2 Lon

PROCESSING PROCESS:

- Import the dataset into Microsoft Power BI Desktop.
- Perform data cleaning using Power Query.
- Remove duplicate and unnecessary data.
- Handle missing values and correct data types.
- Transform and organize the dataset for analysis.
- Create calculated measures using DAX
- Organize the data for analysis and visualization

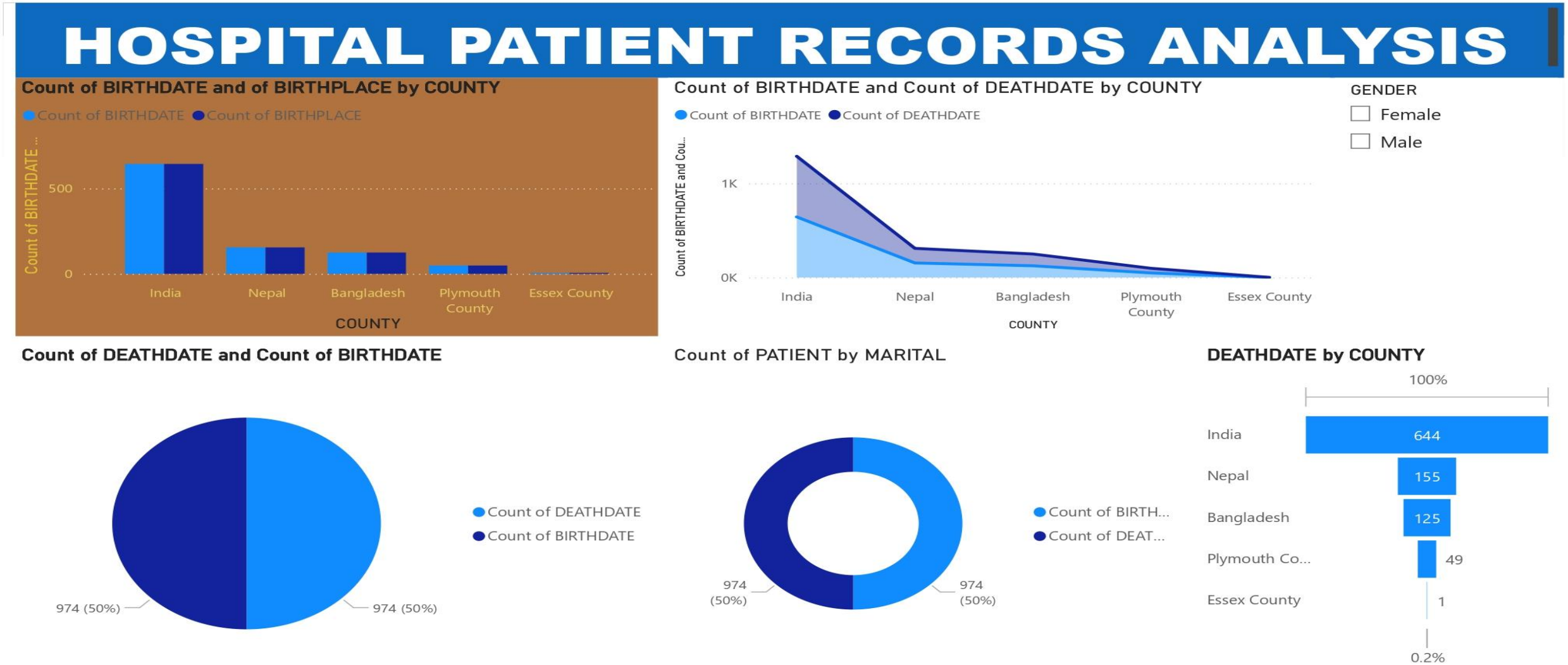
INSIGHT GENERATION:

Visualizations such as KPI cards, line charts, donut charts, and bar charts are used to display production trends, seasonal distribution, and top producing crops or states.

DEPLOYMENT PROCESS:

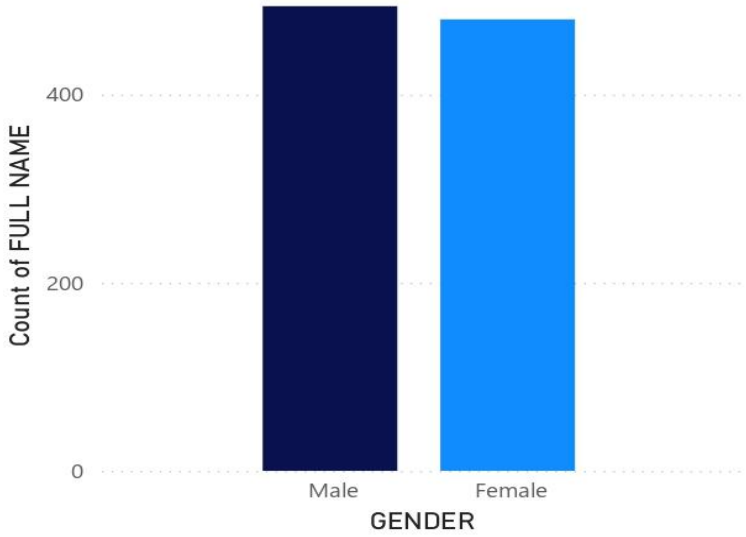
The final dashboard is created and presented with the use of Microsoft Power BI. The dashboard enables users to interact with the data by filtering and slicing to gain insights on crop production based on various regions, seasons, and years.

RESULT:


HOSPITAL PATIENT RECORDS ANALYSIS
Count of BIRTHDATE and of BIRTHPLACE by COUNTY
Count of BIRTHDATE
Count of BIRTHPLACE
Count of BIRTHDATE ...
500
0
India
Nepal
Bangladesh
Plymouth County
Essex County
COUNTY
Count of BIRTHDATE and Count of DEATHDATE by COUNTY
Count of BIRTHDATE
Count of DEATHDATE
Count of BIRTHDATE and Cou...
1K
0K
India
Nepal
Bangladesh
Plymouth County
Essex County
COUNTY
GENDER
Female
Male
Count of DEATHDATE and Count of BIRTHDATE
Count of DEATHDATE
Count of BIRTHDATE
974 (50%)
974 (50%)
Count of PATIENT by MARITAL
Count of BIRTH...
Count of DEAT...
974 (50%)
974 (50%)
DEATHDATE by COUNTY
100%
India
644
Nepal
155
Bangladesh
125
Plymouth Co...
49
Essex County
1
0.2%

Microsoft Elevate

Count of FULL NAME by GENDER



Female

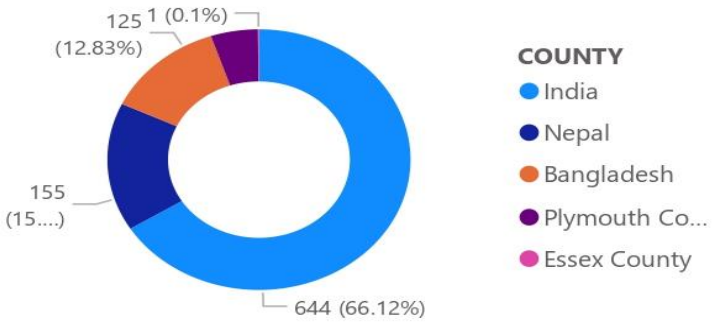
480
Count of FULL NAME

Male

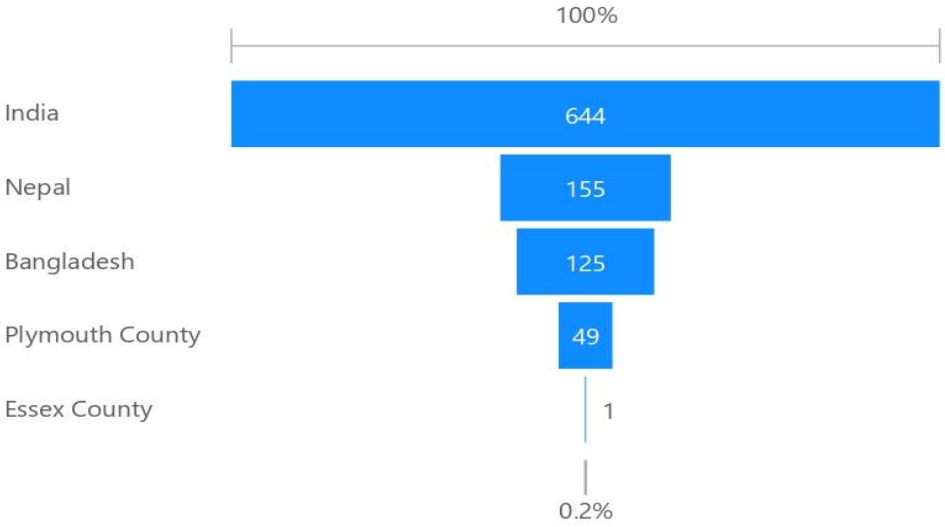
494
Count of FULL NAME

GENDER
- Female
- Male

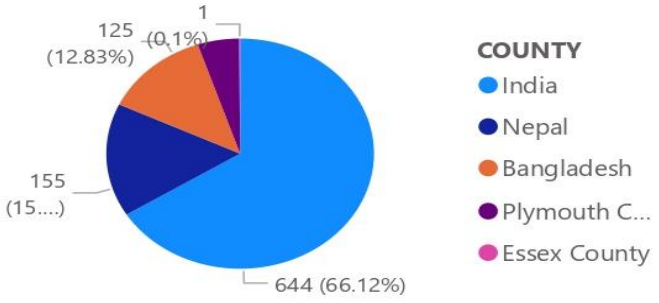
Count of GENDER by COUNTY



Count of GENDER by COUNTY



Count of GENDER by COUNTY



CONCLUSION:

The purpose of the project is to show how the data of patients can be effectively analysed and visualized using the Microsoft Power BI desktop application. By converting the raw data into a more presentable and interactive form, complex data becomes easier to analyse and understand. The data visualizations developed as part of the project enable the user to easily identify the trends of the patients' demographics, such as the gender, marital status, race, ethnicity, and geographical location. This way, the user can easily analyse the data and obtain useful insights from the large amount of data. Thus, the project demonstrates the importance of data visualization, as it enables the user to easily analyse the complex data and obtain a better understanding of the data. Overall, the project provides a clear and effective way of presenting the patients' data.

FUTURE SCOPE:

The future scope of the project is to extend the dashboard with more detailed information about healthcare, such as the medical history of the patients, the number of visits made by the patients to the hospitals, and the treatments provided to the patients. Advanced features of analytics can be incorporated using the Microsoft Power BI Desktop application for further analysis of the patients. Real-time integration of the data can also be incorporated with the help of the application. The dashboard can be further enhanced by adding more interactive features. This will enable the healthcare organizations to gain more insights about the patients.

REFERENCES:

1. https://mavenanalytics.io/data-playground

2. GitHub Link: https://github.com/laithumraja51-afk/Hospital-.git

Thank You